

Lecturer: dr hab. inż. Bogdan Trawiński, prof. uczelni

IT Project Management Support

Lecture 5

Work Breakdown Structure

Work Breakdown Structure

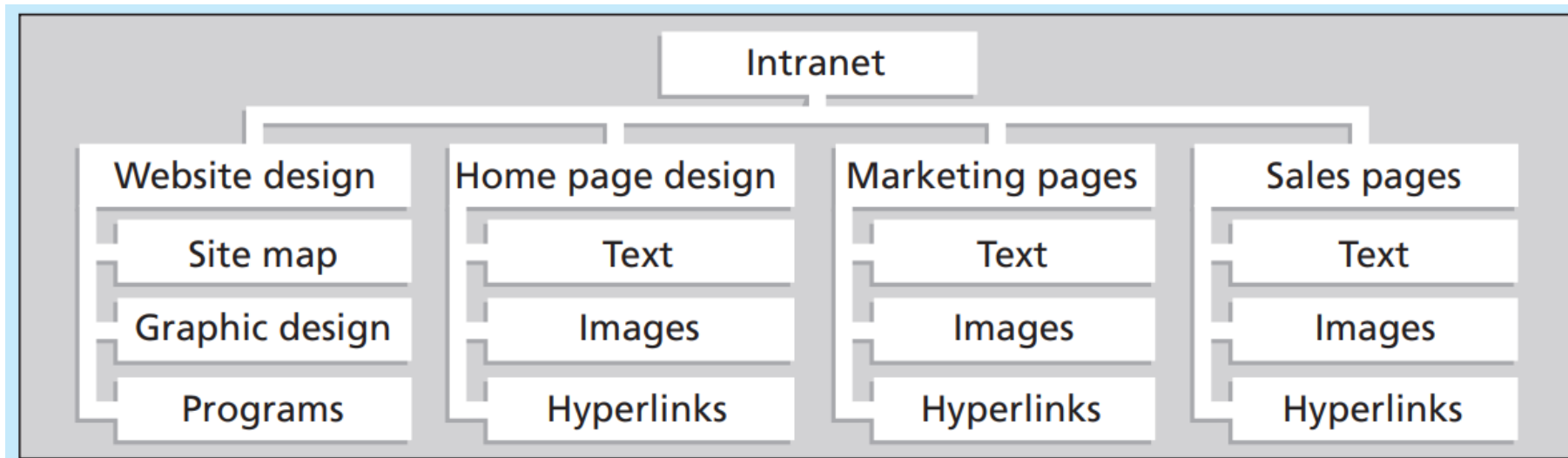
- Work Breakdown Structure (WBS) is a deliverable-oriented grouping of the work involved in a project that defines its total scope.
- Because most projects involve many people and many different deliverables, it is important to organize and divide the work into logical parts based on how the work will be performed.
- The WBS is a foundation document in project management because it provides the basis for planning and managing project schedules, costs, resources, and changes.
- Because the WBS defines the total scope of the project, some project management experts believe that work should not be done on a project if it is not included in the WBS.
- Therefore, it is crucial to develop a complete WBS.

Work Breakdown Structure

- The project scope management plan, scope statement, requirements documentation, enterprise environmental factors, and organizational process assets are the primary inputs for creating a WBS.
- The main tool or technique is decomposition—that is, subdividing project deliverables into smaller pieces.
- The outputs of the process of creating the WBS are the scope baseline and project documents updates.
- The scope baseline includes the approved project scope statement and its associated WBS and WBS dictionary.

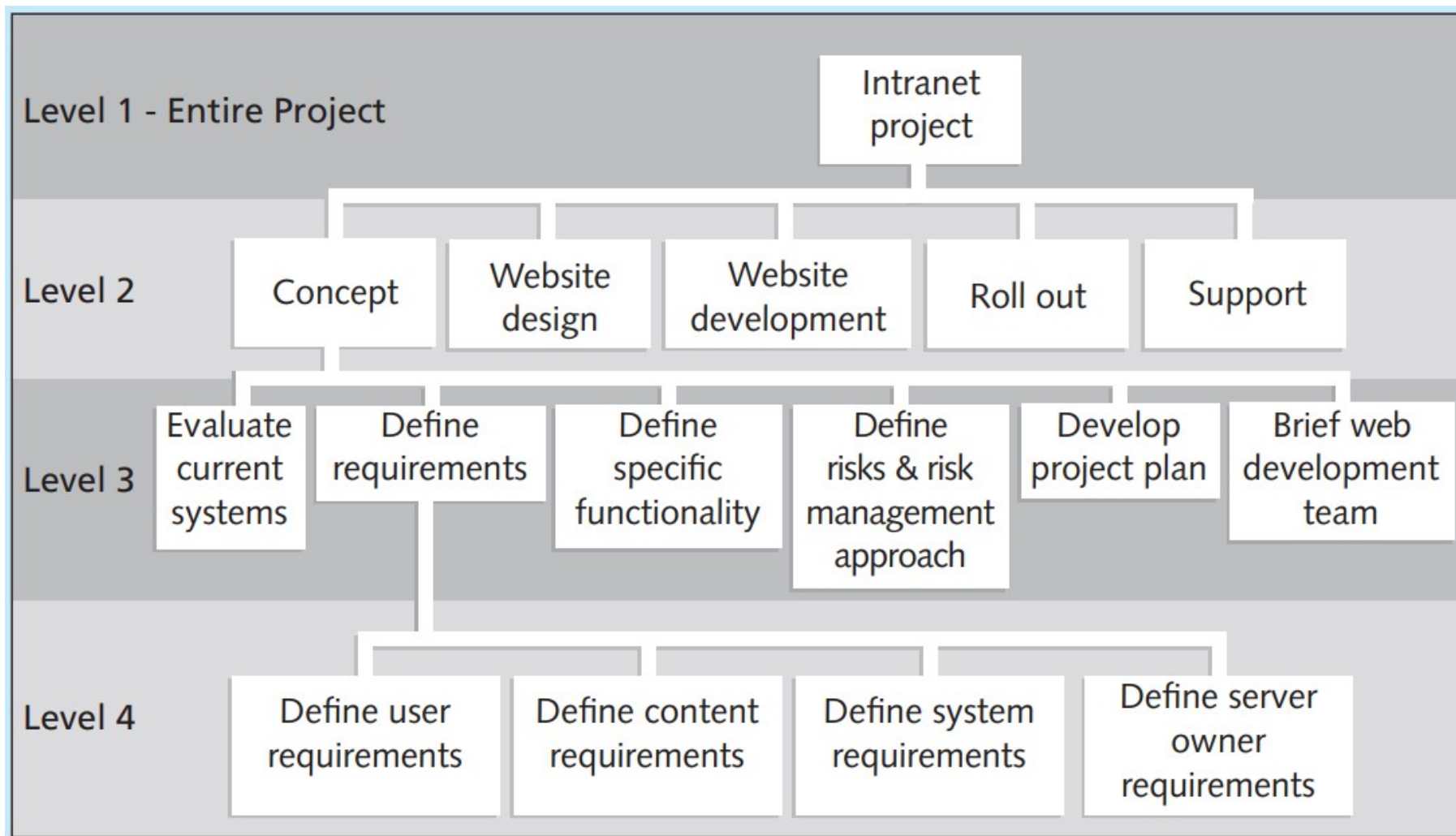
Work Breakdown Structure

- What does a WBS look like?
- A WBS is often depicted as a task-oriented tree of activities, similar to an organizational chart.
- A project team often organizes the WBS around project products, project phases, or the project management process groups.



WBS for an intranet project organized by product

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WBS for an intranet project organized by phase



Tabular form with PMI numbering

1.1 Concept

1.1.1 Evaluate current systems

1.1.2 Define requirements

1.1.2.1 Define user requirements

1.1.2.2 Define content requirements

1.1.2.3 Define system requirements

1.1.2.4 Define server owner requirements

1.1.3 Define specific functionality

1.1.4 Define risks and risk management approach

1.1.5 Develop project plan

1.1.6 Brief web development team

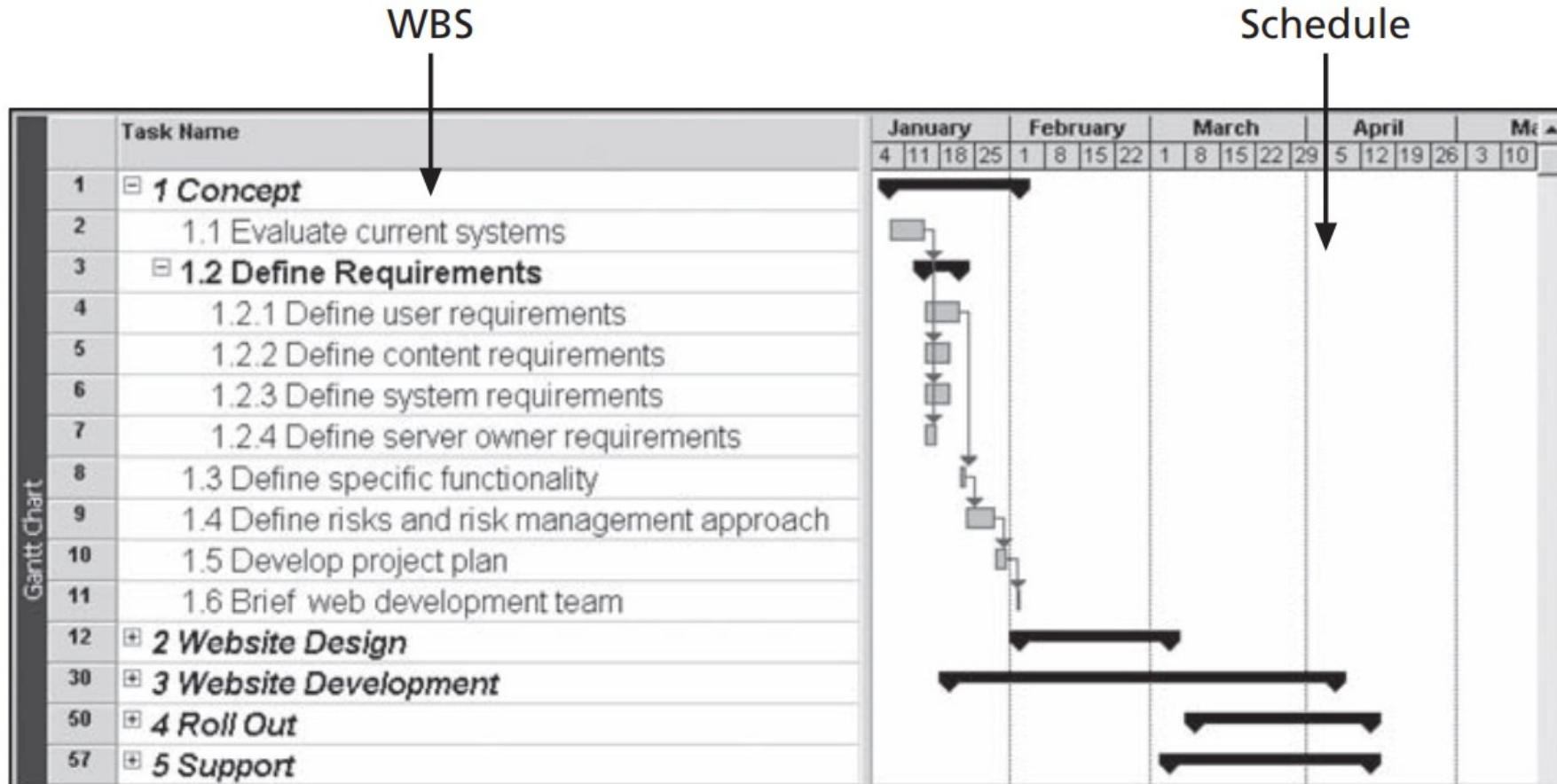
1.2 Website design

1.3 Website development

1.4 Roll out

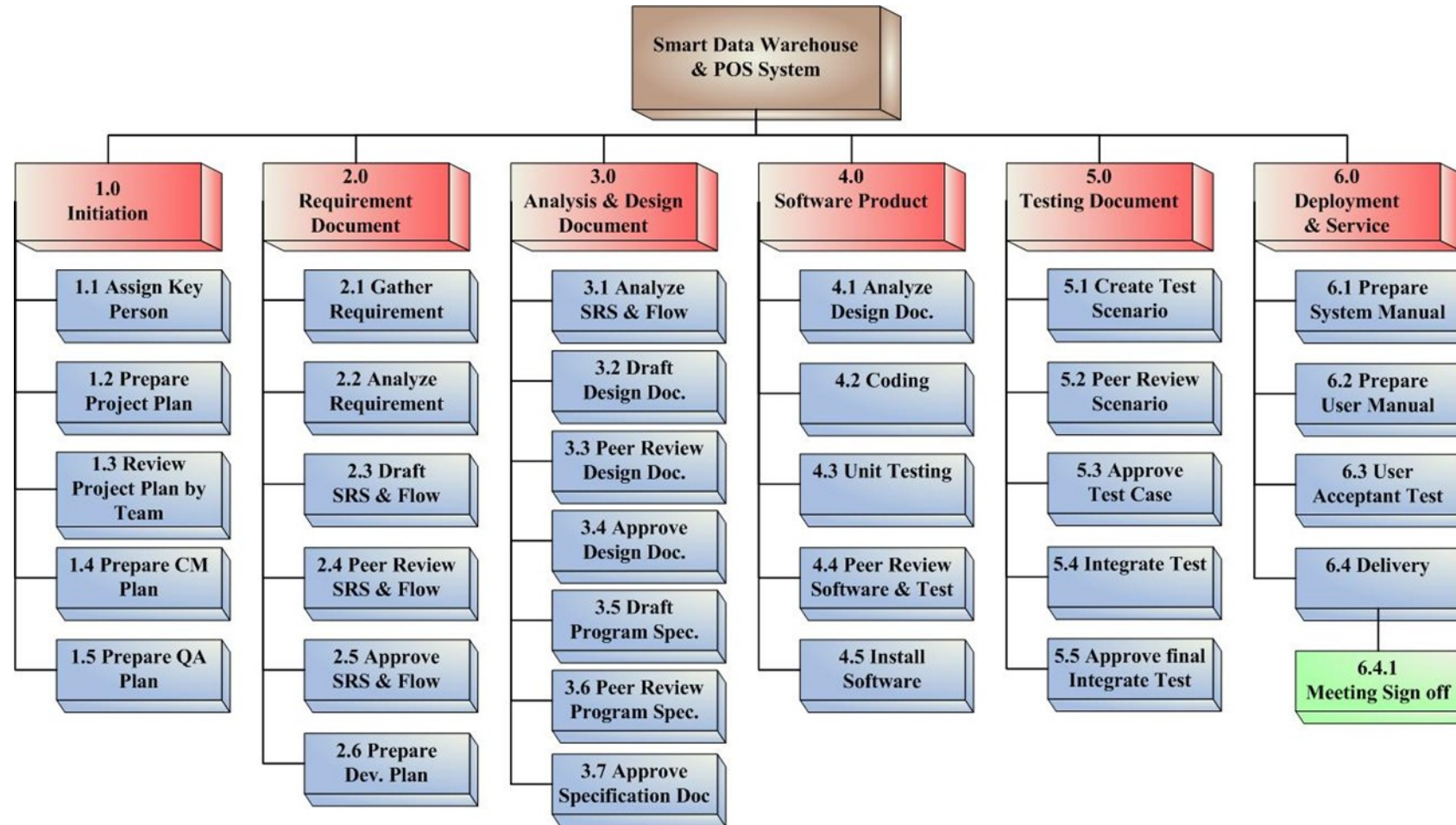
1.5 Support

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WBS and Gantt Chart for the intranet project organized by phase in Ms Project

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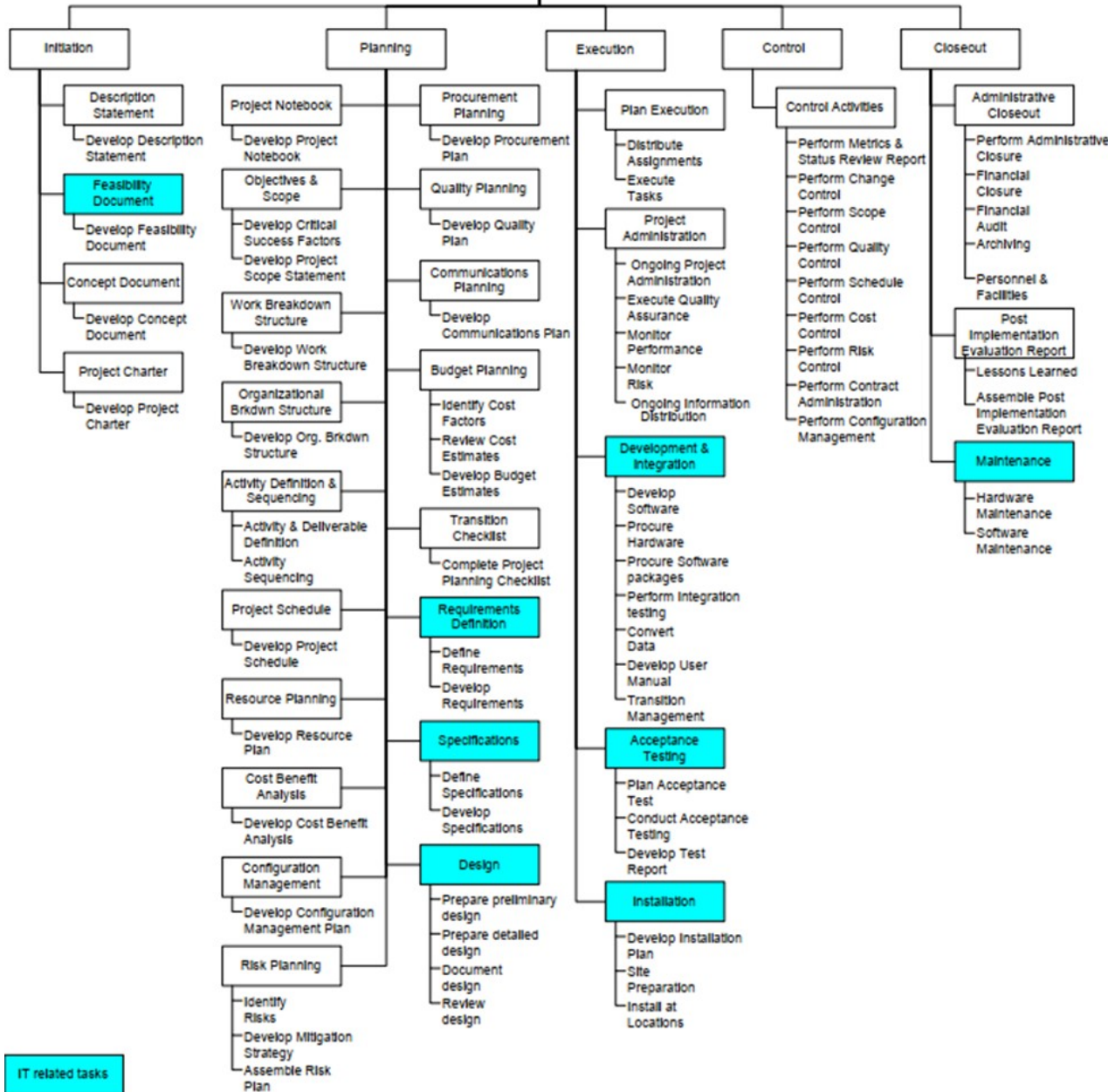
Example of WBS Chart

Work Breakdown Structure (WBS)

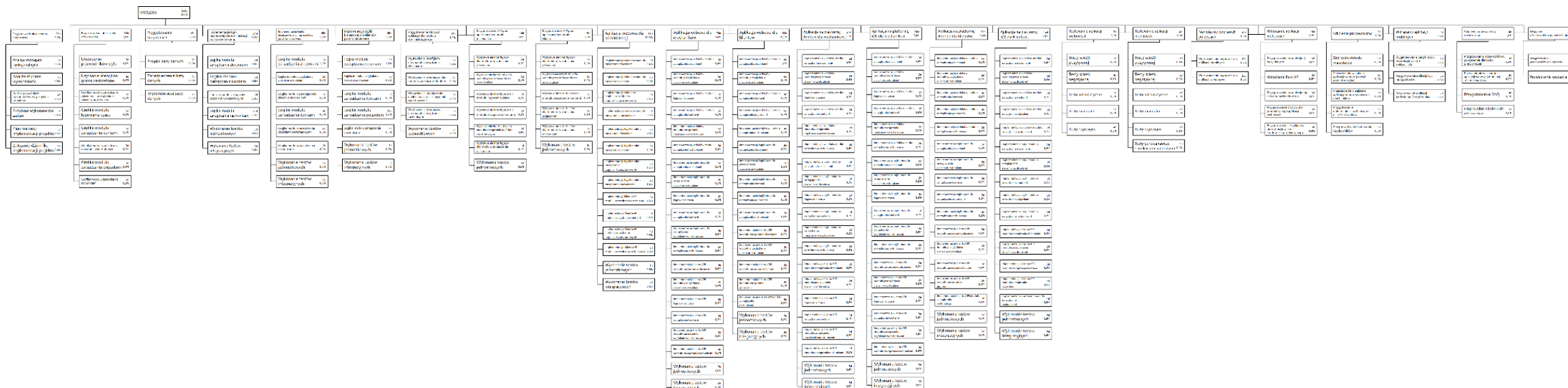
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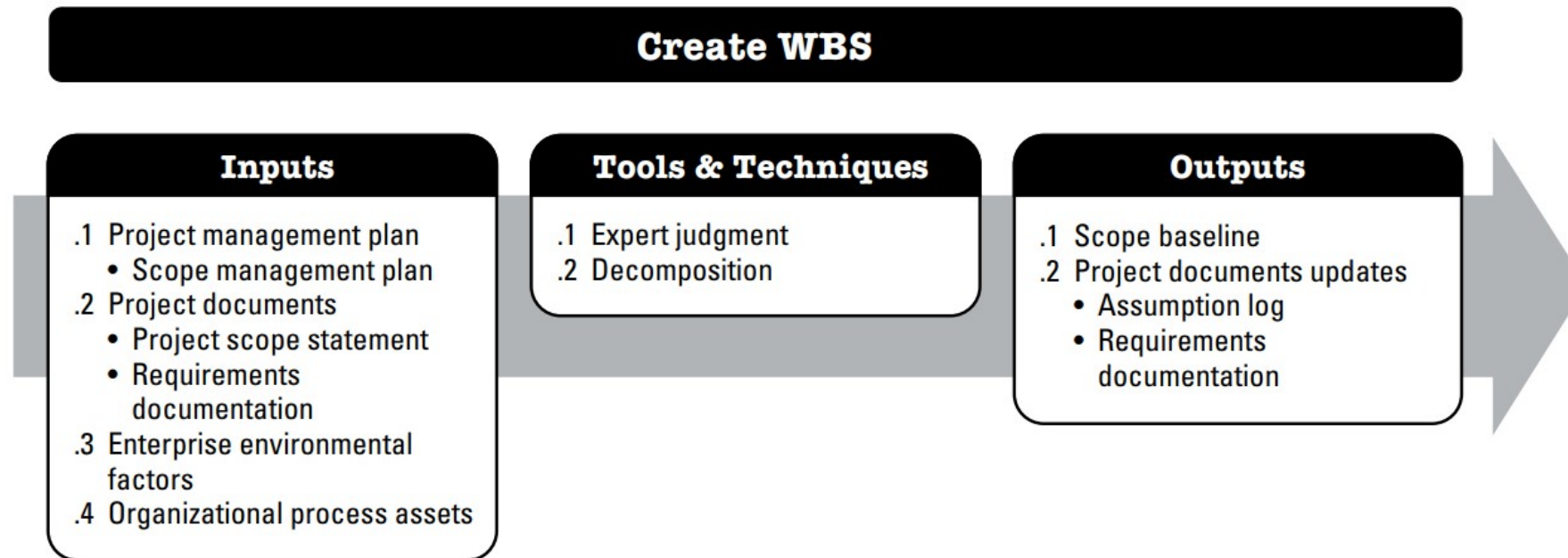


WBS organized by
project
management
proces groups

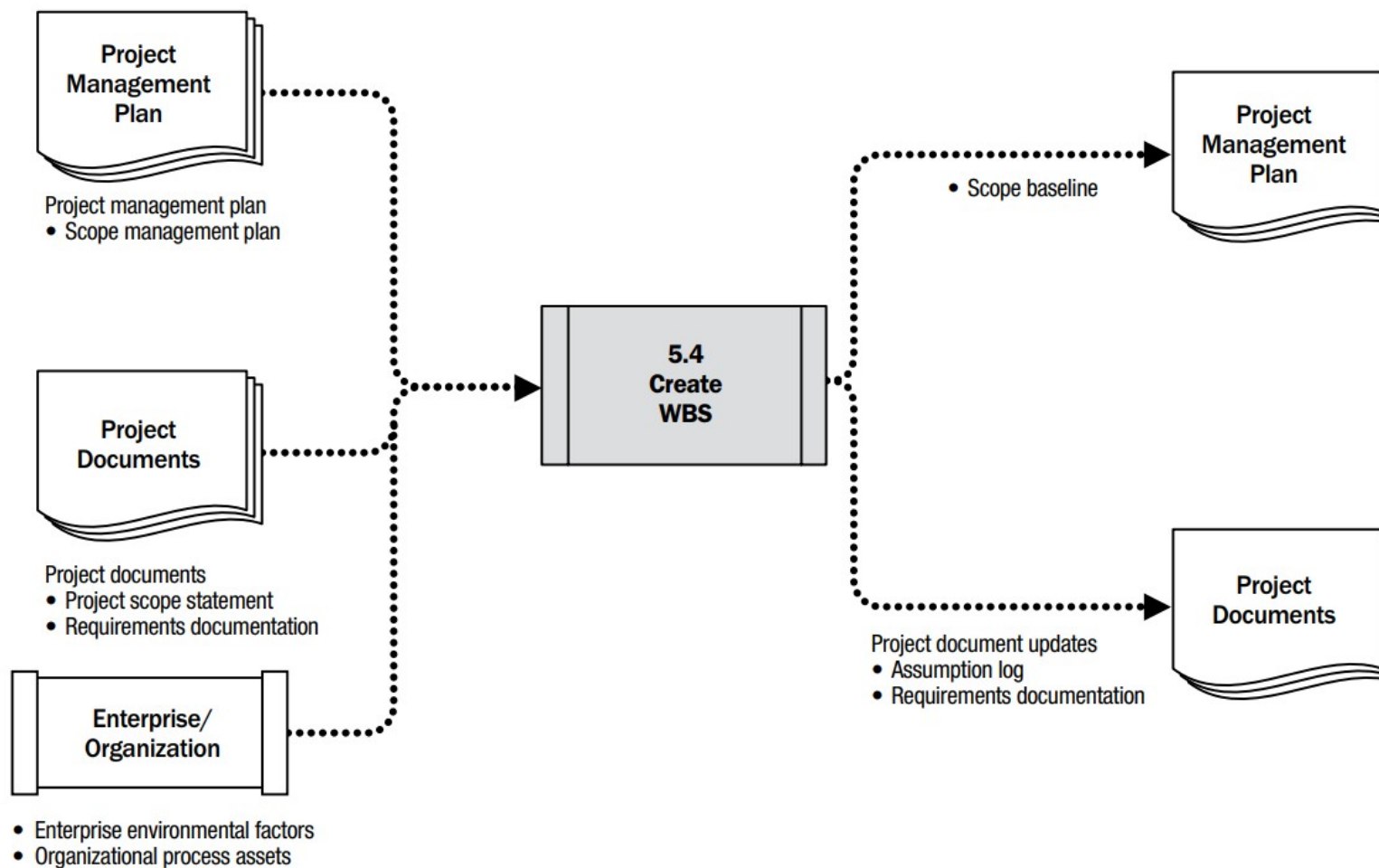


The proces of Creating WBS

- Create WBS is the process of subdividing project deliverables and project work into smaller, more manageable components.
- The key benefit of this process is that it provides a framework of what has to be delivered.



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Work Breakdown Structure

- The WBS is a hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project objectives and create the required deliverables.
- The WBS organizes and defines the total scope of the project and represents the work specified in the current approved project scope statement.
- The planned work is contained within the lowest level of WBS components, which are called **work packages**.
- A **work package** can be used to group the activities where work is scheduled and estimated, monitored, and controlled.
- In the context of the WBS, work refers to work products or deliverables that are the result of activity and not to the activity itself.

WBS Dictionary

- A WBS dictionary is a document that provides detailed information about each WBS item.
- The WBS dictionary is a definition of the work involved in the task - a clarification that makes the summary description in the WBS easier to understand in terms of the approach taken to complete the work.
- The format of the WBS dictionary can vary based on project needs. It might be appropriate to have a short paragraph describing each work package.
- For a more complex project, an entire page or more might be needed for each of the work package descriptions.
- Some projects might require that each WBS item describe the responsible organization, resource requirements, estimated costs, dependencies with other activities, and other information.
- Project teams often review WBS dictionary entries from similar tasks to get a better idea of how to create their entries.

WBS Dictionary (according to PMBOK Guide, ch. 5, p. 162)

◆ **WBS dictionary.** The WBS dictionary is a document that provides detailed deliverable, activity, and scheduling information about each component in the WBS. The WBS dictionary is a document that supports the WBS. Most of the information included in the WBS dictionary is created by other processes and added to this document at a later stage. Information in the WBS dictionary may include but is not limited to:

- Code of account identifier,
- Description of work,
- Assumptions and constraints,
- Responsible organization,
- Schedule milestones,
- Associated schedule activities,
- Resources required,
- Cost estimates,
- Quality requirements,
- Acceptance criteria,
- Technical references, and
- Agreement information.

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WBS Dictionary Entry March 20

Project Title: Information Technology (IT) Upgrade Project

WBS Item Number: 2.2

WBS Item Name: Update Database

Description: The IT department maintains an online database of hardware and software on the corporate intranet. We need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Perform Physical Inventory, and must precede WBS Item Number 3.0, Acquire Hardware and Software.

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WBS Dictionary template

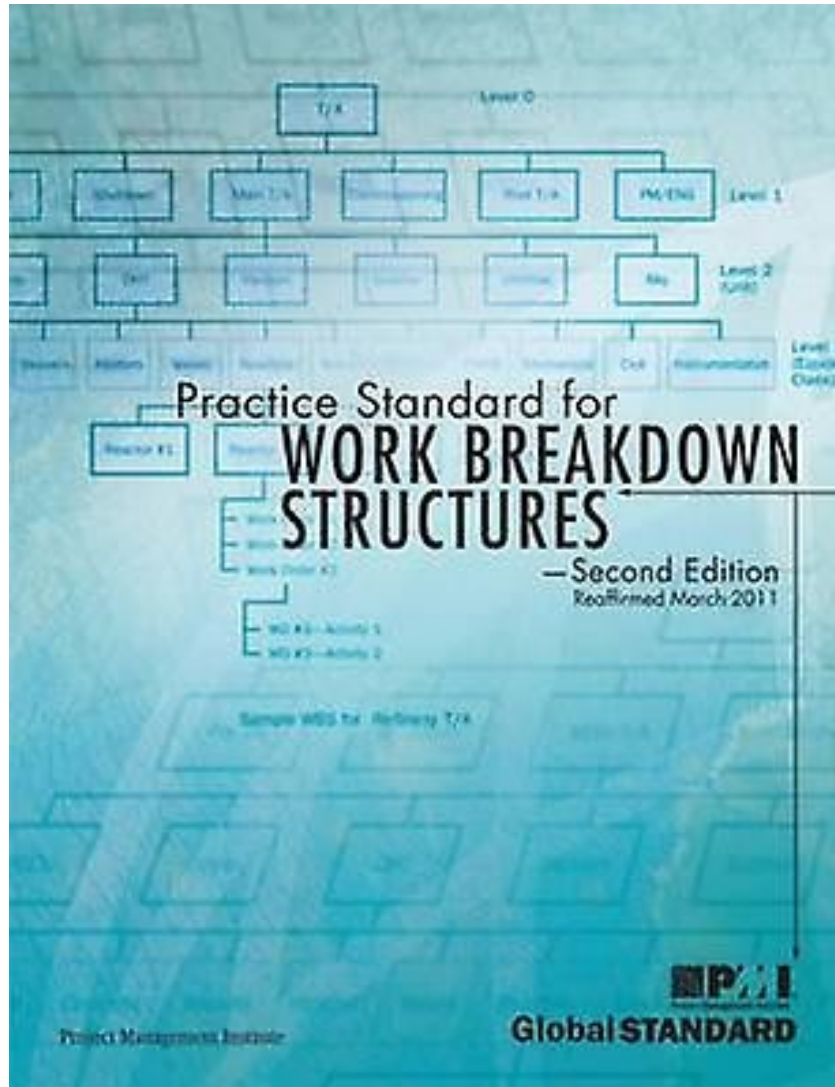
No.	Item	Description
1	WBS Code	Enter the WBS Title and identify it as a Work Package, Planning and/or Control Account.
2	Responsible Organization/ Individual	Name the single organization, group or individual that has been assigned sole responsibility for making sure the Work Package is completed. Include contact information.
3	Description	Define Work Package boundaries. Frame the scope content so that it is clear what scope is included and excluded. If it is a Planning Package, describe the known and unknown scope.
4	Deliverables	Identify the product, service or results created when all of the work in this Work Package is complete. Include any critical intermediate deliverables.
5	Acceptance Criteria	Describe the functional and physical requirements in order to meet customer expectations and quality requirements. Include any unique approvals required for acceptance.
6	Budget	Designate the budget for this Work Package, plus any critical resource information and assumptions.
7	Milestones	List any Start Dates, End Dates, Intermediate Milestones, Interdependencies, Constraints and any assumption for the deliverables.
8	Risks	Include any known threats and opportunities with response strategies.
9	Additional Information	Describe any additional information, such as references, related work packages, etc.
10	Approvals:	Date: Rev:

Approaches to Developing Work Breakdown Structures

You can use several approaches to develop a work breakdown structure:

- Using guidelines
- The analogy approach
- The top-down approach
- The bottom-up approach
- The mind-mapping approach

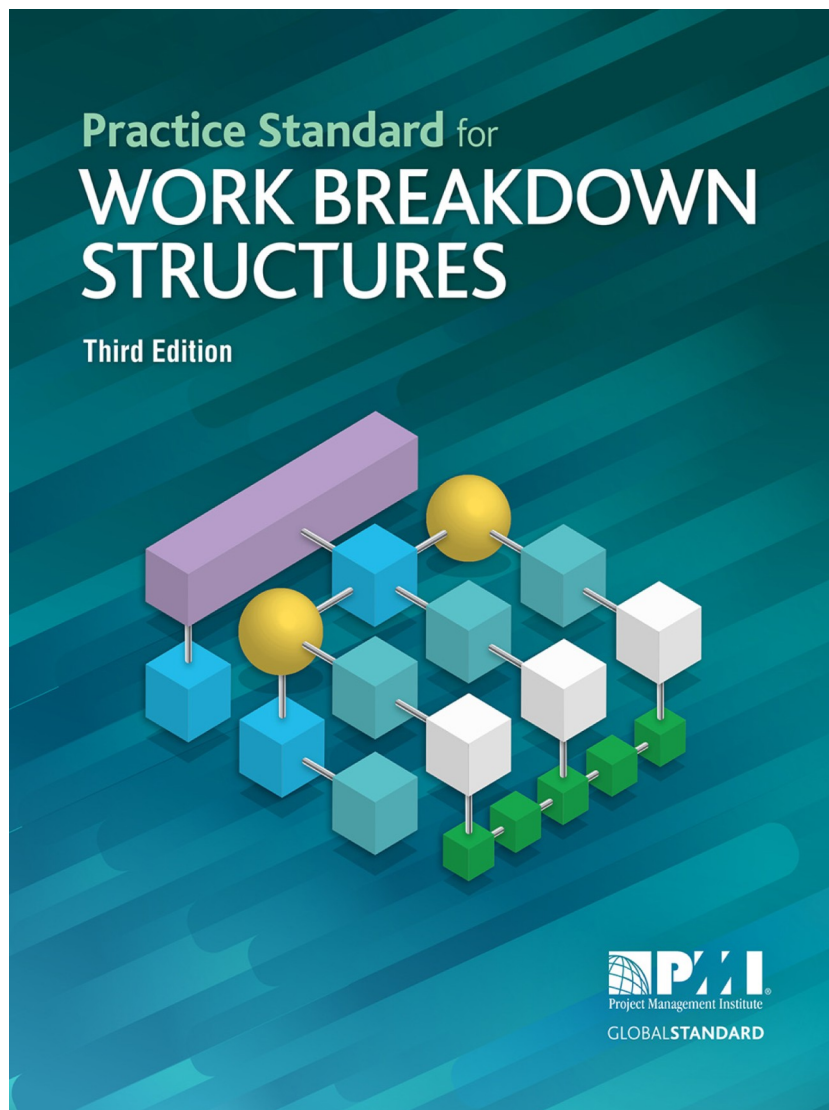
The description you can find in the ITPM Textbook, ch. 5



Practice Standard for Work Breakdown Structures – Second Edition (2006)

Work breakdown structures (WBS) are a guide for defining work as it relates to a specific project's objectives. Project professionals use a WBS to define project deliverables and establish the structure to manage work to completion. The Practice Standard for Work Breakdown Structures – Second Edition (Reaffirmed) provides direction for the preliminary development and implementation of the WBS.

Throughout this standard, you will learn what characterizes a high-quality WBS and discover the substantial benefits of using the WBS in everyday, real-life situations.



Practice Standard for Work Breakdown Structures – Third Edition (2019)

As a comprehensive update to the Practice Standard for Work Breakdown Structures – Second Edition, the Third Edition applies the WBS to agile, iterative, predictive and incremental project life cycles, while also exploring several different types of decomposition in practice today.

This practice standard aligns with other recent PMI standards, including A Guide to the Project Management Body of Knowledge (PMBOK® Guide) – Sixth Edition, providing practitioners with an indispensable tool to consistently track schedule, budget, risk and performance, regardless of discipline.

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WBS Creation Method	Advantages	Challenges
Top-Down	• Structures project conveniently for status reporting • Helps ensure projects are logically structured • Is valuable when brainstorming/ discovering project deliverables • Can accommodate additional deliverables as they are uncovered	• Requires constant attention that no work packages are overlooked • WBS needs to be elaborated to sufficiently detailed level to permit management oversight and control
Bottom-Up	• Starts with all deliverables and works backwards into a project • Confirms that all work packages are included	• Identifying all deliverables before producing the WBS • Making sure work packages are logically grouped • Can lose focus on big picture
WBS Standards	• Formats are predefined • Enhances cross-project WBS consistency	• Making a project fit the standard • Can lead to inclusion of unnecessary deliverables or failure to include project-specific deliverables • Not all projects fit into a highly structured set of WBS standards
WBS Templates	• Provides a starting point for WBS creation • May help determine appropriate level of detail required • Enhances cross-project WBS consistency	• Requires a project fit the standard • Can lead to inclusion of unnecessary deliverables or failure to include project-specific deliverables • Not all projects fit into a highly structured set of WBS templates

According to Practice Standard for Work Breakdown Structures – Second Edition (2006)

Advice for Creating a WBS and WBS Dictionary (1)

Creating a good WBS is no easy task and usually requires several iterations. Often, it is best to use a combination of approaches to create a project's WBS. Some basic principles, however, apply to creating any good WBS and its WBS dictionary:

- A unit of work should appear in only one place in the WBS.
- The work content of a WBS item is the sum of the WBS items below it.
- A WBS item is the responsibility of only one person, even though many people might be working on it.
- The WBS must be consistent with the way work actually will be performed; it should serve the project team first, and serve other purposes only if practical.

Advice for Creating a WBS and WBS Dictionary (2)

- Project team members should be involved in developing the WBS to ensure consistency and buy-in.
- Each WBS item must be documented in a WBS dictionary to ensure accurate understanding of the scope of work included and not included in that item.
- The WBS must be a flexible tool to accommodate inevitable changes while properly maintaining control of the work content in the project according to the scope statement.

WBS design principles

- 100% rule
- Mutually exclusive elements
- Plan outcomes, not actions
- Level of detail
- Coding scheme
- Terminal element
- Consistent to norms

The 100% Rule

One of the most important WBS design principles is called the 100% Rule. The *Practice Standard for Work Breakdown Structures (Second Edition)*, published by the Project Management Institute (PMI) defines the 100% Rule as follows:

The 100% Rule...states that the WBS includes 100% of the work defined by the project scope and captures all deliverables – internal, external, interim – in terms of the work to be completed, including project management.

- *The 100% rule is one of the most important principles guiding the development, decomposition and evaluation of the WBS.*
- *The rule applies at all levels within the hierarchy: the sum of the work at the “child” level must equal 100% of the work represented by the “parent” and the WBS should not include any work that falls outside the actual scope of the project, that is, it cannot include more than 100% of the work...*
- *It is important to remember that the 100% rule also applies to the activity level. The work represented by the activities in each work package must add up to 100% of the work necessary to complete the work package.*

Mutually exclusive elements

- Mutually exclusive: In addition to the 100% rule, it is important that there is no overlap in scope definition between different elements of a work breakdown structure.
- This ambiguity could result in duplicated work or miscommunications about responsibility and authority.
- Such overlap could also cause confusion regarding project cost accounting.
- If the WBS element names are ambiguous, a WBS dictionary can help clarify the distinctions between WBS elements.
- The WBS Dictionary describes each component of the WBS with milestones, deliverables, activities, scope, and sometimes dates, resources, costs, quality.

Plan outcomes, not actions

- If the work breakdown structure designer attempts to capture any action-oriented details in the WBS, the designer will likely include either too many actions or too few actions.
- Too many actions will exceed 100% of the parent's scope and too few will fall short of 100% of the parent's scope. The best way to adhere to the 100% rule is to define WBS elements in terms of outcomes or results, not actions.
- This also ensures that the WBS is not overly prescriptive of methods, allowing for greater ingenuity and creative thinking on the part of the project participants.
- Work breakdown structures that subdivide work by project phases must ensure that phases are clearly separated by a deliverable also used in defining entry and exit criteria.

Level of detail (1)

- One must decide when to stop dividing work into smaller elements.
- For most projects a hierarchy of two to four levels will suffice.
- This will assist in determining the duration of activities necessary to produce a deliverable defined by the WBS.
- There are several heuristics or "rules of thumb" used when determining the appropriate duration of an activity or group of activities necessary to produce a specific deliverable defined by the WBS.

Level of detail (2)

- The first is the "80 hour rule" which means that no single activity or group of activities at the lowest level of detail of the WBS to produce a single deliverable should be more than 80 hours of effort.
- The second rule of thumb is that no activity or group of activities at the lowest level of detail of the WBS should be longer than a single reporting period. Thus if the project team is reporting progress monthly, then no single activity or series of activities should be longer than one month long.
- The last heuristic is the "if it makes sense" rule. Applying this rule of thumb, one can apply "common sense" when creating the duration of a single activity or group of activities necessary to produce a deliverable defined by the WBS.

Coding scheme

- It is common for work breakdown structure elements to be numbered sequentially to reveal the hierarchical structure.
- The purpose for the numbering is to provide a consistent approach to identifying and managing the WBS across like systems regardless of vendor or service
- A coding scheme also helps WBS elements to be recognized in any written context and allows for mapping to the WBS Dictionary.



Tabular form with PMI numbering

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1.4 Roll out

1.5 Support

Terminal element

- Terminal element is the lowest element in a tree structure, is one that is not further subdivided.
- In a Work Breakdown Structure such elements (activity or deliverable), also known as work packages, are the items that are estimated in terms of resource requirements, budget and duration; linked by dependencies; and scheduled.
- A WBS can be expressed down to any level of interest.
- Three levels are the minimum recommended, with additional levels for and only for items of high cost or high risk and two levels of detail at cases such as systems engineering or program management, with the standard showing examples of WBS with **varying depth such as software development at points going to 5 levels.**

Consistent to norms

- The higher WBS structure should be consistent to whatever norms or template mandates exist within the organization or domain.
- For example, shipbuilding for the U.S. Navy must respect that the nautical terms and their hierarchy structure put into MIL-STD are embedded in Naval Architecture and that matching Navy offices and procedures have been built to match this naval architecture structure, so any significant change of WBS element numbering or naming in the hierarchy would be unacceptable.

What are the Top 20 Benefits of Using the WBS?

1. Understand work at early stages
2. Avoid uncontrolled changes
3. Deliver what is expected
4. Understand areas with limited understanding
5. Visualize internal and external work
6. Visualize project boundaries and manage complexities
7. Provide a baseline for scope change control
8. Assign and explain the work
9. Enhance project planning
10. Avoid replanning and detect early warnings

What are the Top 20 Benefits of Using the WBS?

11. Build strong foundations for acquisitions
12. Improve communications
13. Achieve a common understanding of the work
14. Improve project reporting
15. Gain stakeholders' support
16. Monitor, measure and control work better
17. Inspire confidence and gain credibility
18. Improve future projects
19. Compare scope among projects
20. Integrate scope with time and cost



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Thank You very much
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